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Course: BTech Cse Data Science

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Subject: Statistical Foundation of Data
Science

Explanation Of Code :

I am using Pandas and NumPy to work with student data and perform some mathematical operations on arrays.”

“First, I import NumPy as np and Pandas as pd. NumPy is mainly used for numerical and matrix operations, while Pandas is used to handle tabular data.”

“Then I create a dictionary called data. It contains the students' names, ages, marks in Math and Python, and their departments.”

“After that, I convert this dictionary into a Pandas DataFrame using `pd.DataFrame(data)`. A DataFrame is basically a table with rows and columns, similar to an Excel sheet.”

“Next, I select only the numerical columns — Age, Math and Python — and convert them into a NumPy array using `.to_numpy()`. I do this because NumPy is useful for numerical calculations.”

“Then I check three things: size, shape and dimensions. Size tells me the total number of elements, shape tells me the number of rows and columns, and dimensions tells me whether the array is one-dimensional, two-dimensional, etc.”

“After that, I create two matrices, A and B, and perform different matrix operations on them: addition, subtraction, multiplication, dot product, transpose and inverse.”

“Finally, I print the results of all these operations.”

```
import numpy as np
import pandas as pd
```

```
data = {
    "Name": ["Manish", "Rahul", "Priya", "Neha", "Rohit"],
    "Age": [20, 21, 20, 22, 21],
    "Math": [85, 78, 92, 75, 88],
    "Python": [90, 82, 88, 80, 91],
    "Department": ["CSE", "AI", "CSE", "DS", "AI"]
}
```

```
df = pd.DataFrame(data)
```

```
print("RAW DATA:")
print(df)
```

RAW DATA:

	Name	Age	Math	Python	Department
0	Manish	20	85	90	CSE
1	Rahul	21	78	82	AI
2	Priya	20	92	88	CSE
3	Neha	22	75	80	DS
4	Rohit	21	88	91	AI

```
numeric_data = df[["Age", "Math", "Python"]].to_numpy()

print("NUMPY ARRAY:")
print(numeric_data)
```

NUMPY ARRAY:

```
[[20 85 90]
 [21 78 82]
 [20 92 88]
 [22 75 80]
 [21 88 91]]
```

```
print("Size:", numeric_data.size)
print("Shape:", numeric_data.shape)
print("Dimensions:", numeric_data.ndim)
```

Size: 15
Shape: (5, 3)
Dimensions: 2

```
A = np.array([
    [1, 2],
    [3, 4]
])

B = np.array([
    [5, 6],
    [7, 8]
])
```

```
print("Addition:")
print(A + B)
```

```
Addition:  
[[ 6  8]  
 [10 12]]
```

```
print("Subtraction:")  
print(A - B)
```

```
Subtraction:  
[[-4 -4]  
 [-4 -4]]
```

```
print("Multiplication:")  
print(A * B)
```

```
Multiplication:  
[[ 5 12]  
 [21 32]]
```

```
print("Dot Product:")  
print(np.dot(A, B))
```

```
Dot Product:  
[[19 22]  
 [43 50]]
```

```
print("Transpose:")  
print(A.T)
```

```
Transpose:  
[[1 3]  
 [2 4]]
```

```
print("Inverse:")  
print(np.linalg.inv(A))
```

```
Inverse:  
[[-2.   1. ]  
 [ 1.5 -0.5]]
```